Prática de Física dos Dispositivos Eletrônicos

FGA0100

Laboratório-10

Diodo: Chave de Sinais

01/2022

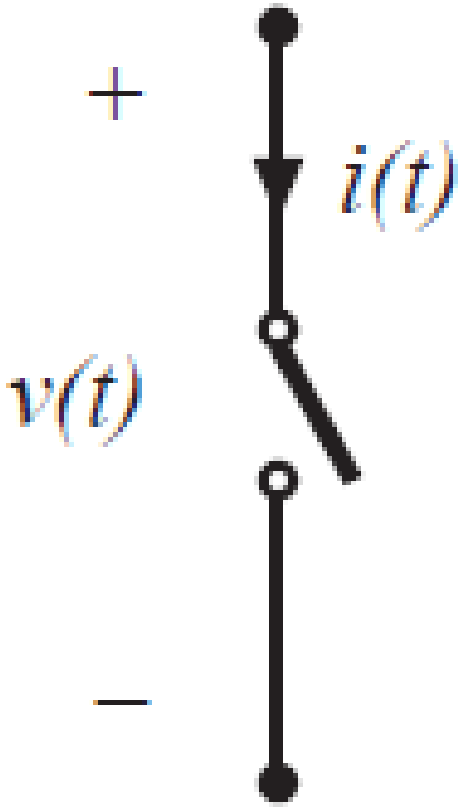
FGA **Universidade de Brasília**

CHAVE IDEAL

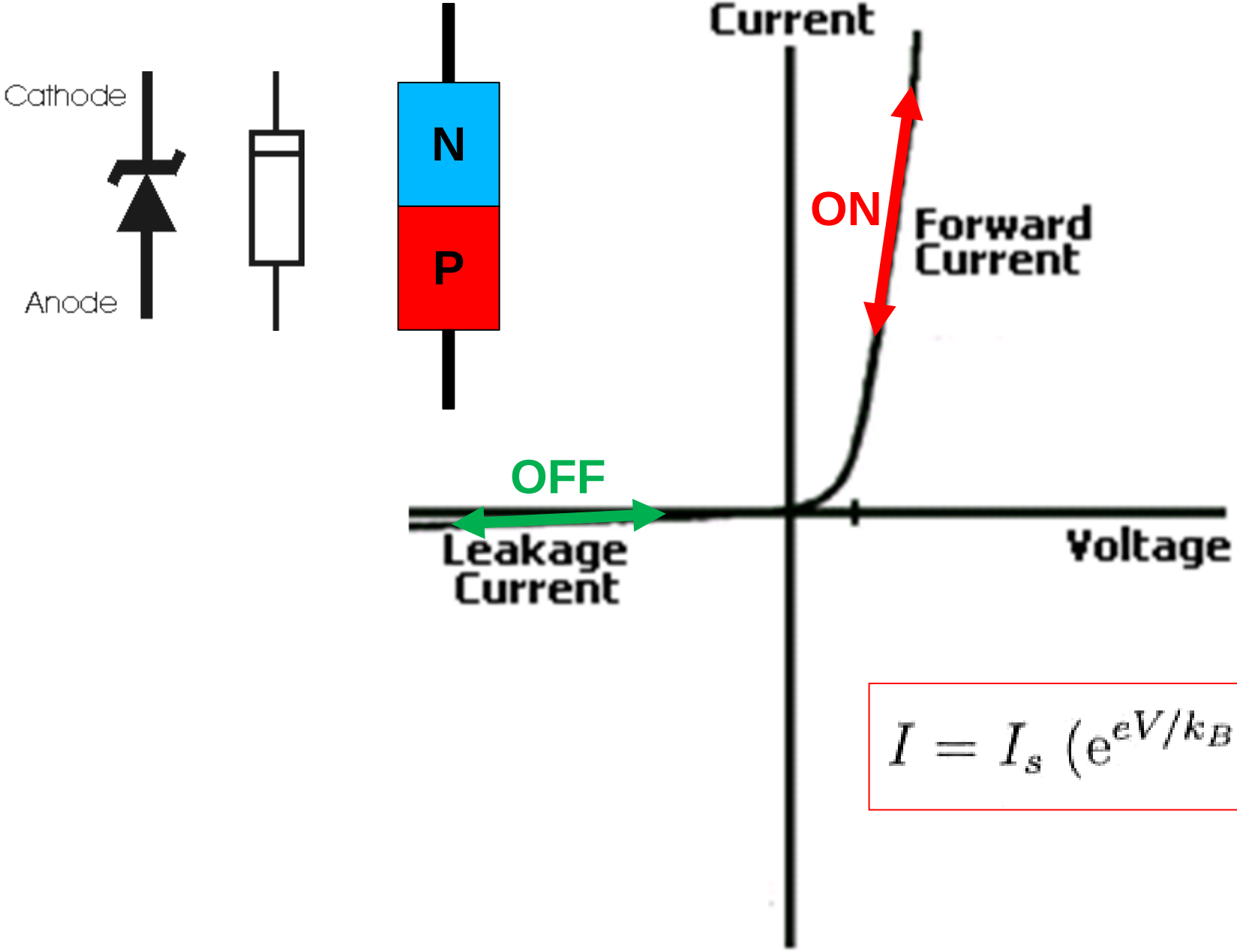
Fechada: $v(t) = 0$

Aberta: $i(t) = 0$

$$p(t) = v(t)\, i(t) = 0$$



Curva I x V do Diodo

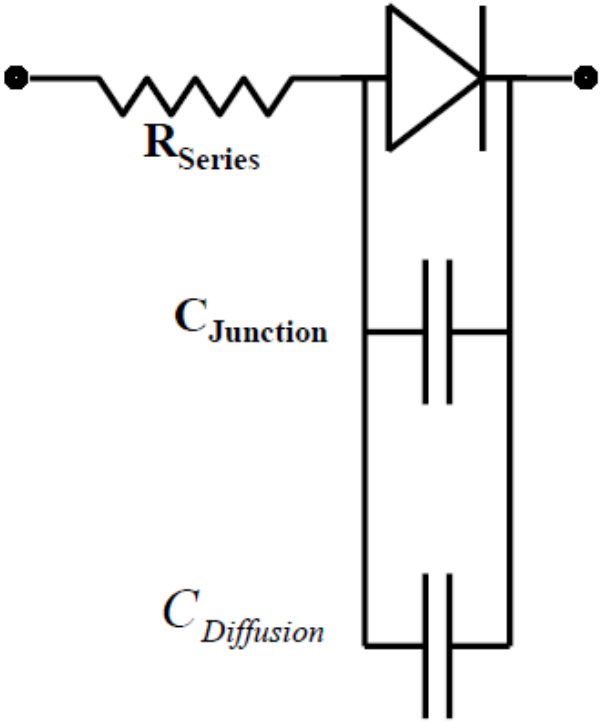


$$I = I_s \left(e^{eV/k_B T} - 1\right)$$

Modelo Não-linear Dinâmico do Diodo

$$I_D = I_s\left[\exp\left(\frac{V}{V_t}\right) - 1\right] \qquad V_t = kT/q$$

$$i_D = I_S\left(e^{(v_D - i_D R_{Series})/\eta V_T} - 1\right)$$



LIMITING VALUES

In accordance with the Absolute Maximum Rating System (IEC 60134).

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
V_{RRM}	repetitive peak reverse voltage		–	100	V
V_R	continuous reverse voltage		–	100	V
I_F	continuous forward current	see Fig.2; note 1	–	200	mA
I_{FRM}	repetitive peak forward current		–	450	mA
I_{FSM}	non-repetitive peak forward current	square wave; T_j = 25 °C prior to surge; see Fig.4			
		t = 1 µs	–	4	A
		t = 1 ms	–	1	A
		t = 1 s	–	0.5	A
P_{tot}	total power dissipation	T_{amb} = 25 °C; note 1	–	500	mW
T_{stg}	storage temperature		–65	+200	°C
T_j	junction temperature		–	200	°C

Note

1. Device mounted on an FR4 printed-circuit board; lead length 10 mm.

ELECTRICAL CHARACTERISTICS

T_j = 25 °C unless otherwise specified.

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
V_F	forward voltage	see Fig.3			
	1N4148	I_F = 10 mA	–	1	V
	1N4448	I_F = 5 mA	0.62	0.72	V
		I_F = 100 mA	–	1	V
I_R	reverse current	V_R = 20 V; see Fig.5		25	nA
		V_R = 20 V; T_j = 150 °C; see Fig.5	–	50	µA
I_R	reverse current; 1N4448	V_R = 20 V; T_j = 100 °C; see Fig.5	–	3	µA
C_d	diode capacitance	f = 1 MHz; V_R = 0 V; see Fig.6	–	4	pF
t_{rr}	reverse recovery time	when switched from I_F = 10 mA to I_R = 60 mA; R_L = 100 Ω; measured at I_R = 1 mA; see Fig.7	–	4	ns
V_{fr}	forward recovery voltage	when switched from I_F = 50 mA; t_r = 20 ns; see Fig.8	–	2.5	V

Montagens Experimentais

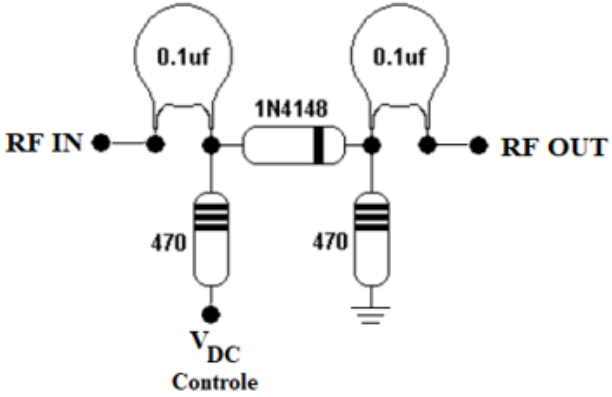


Figura-1

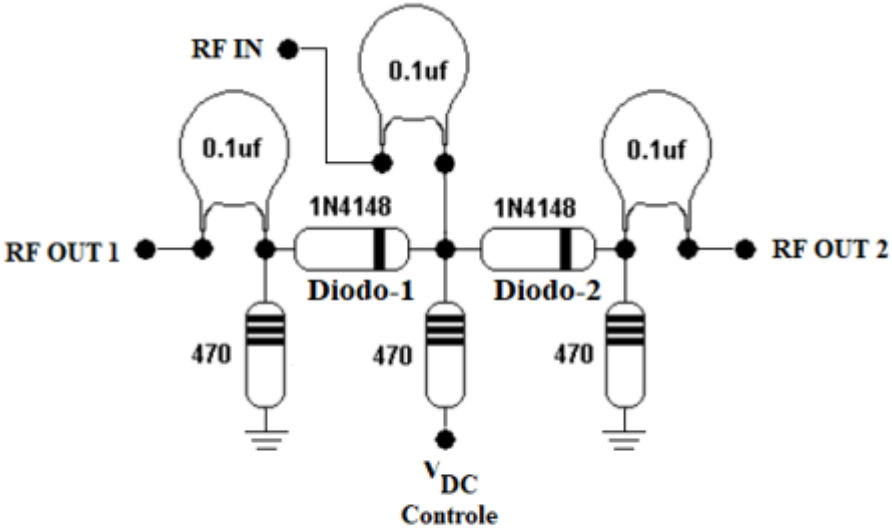
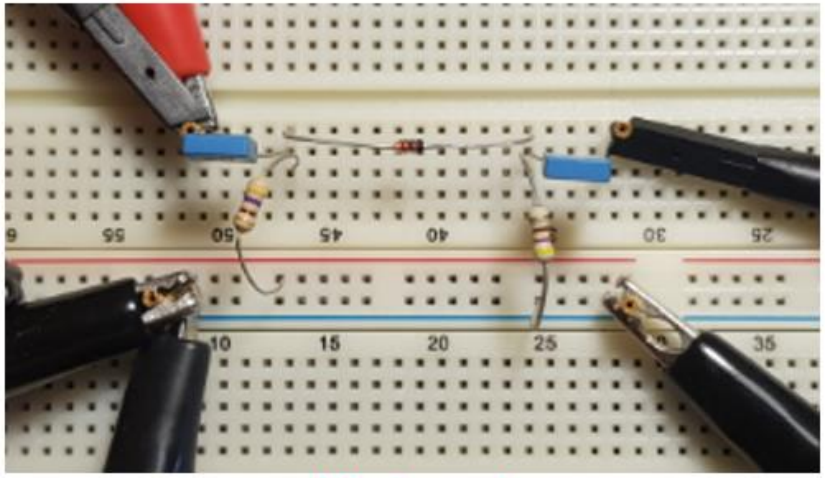
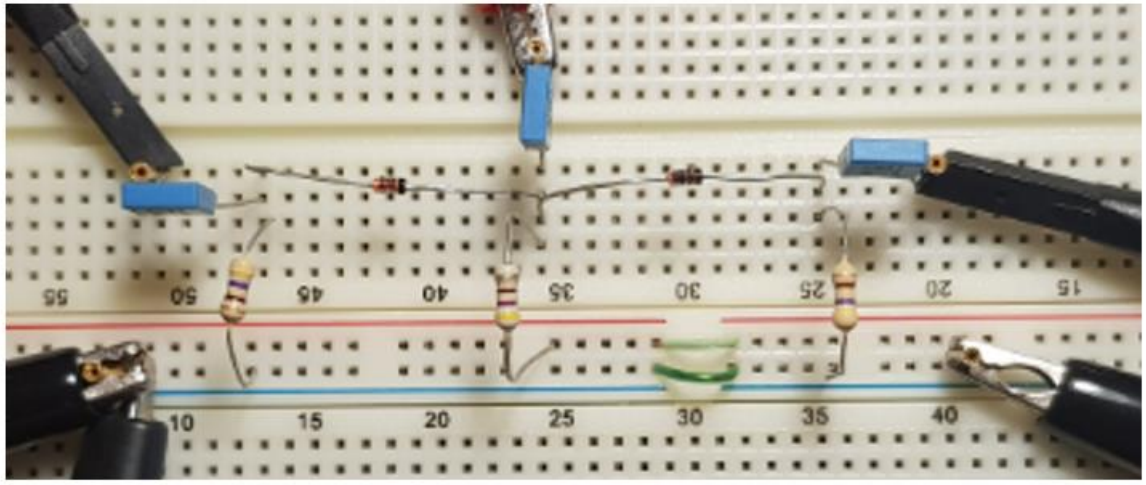


Figura-2

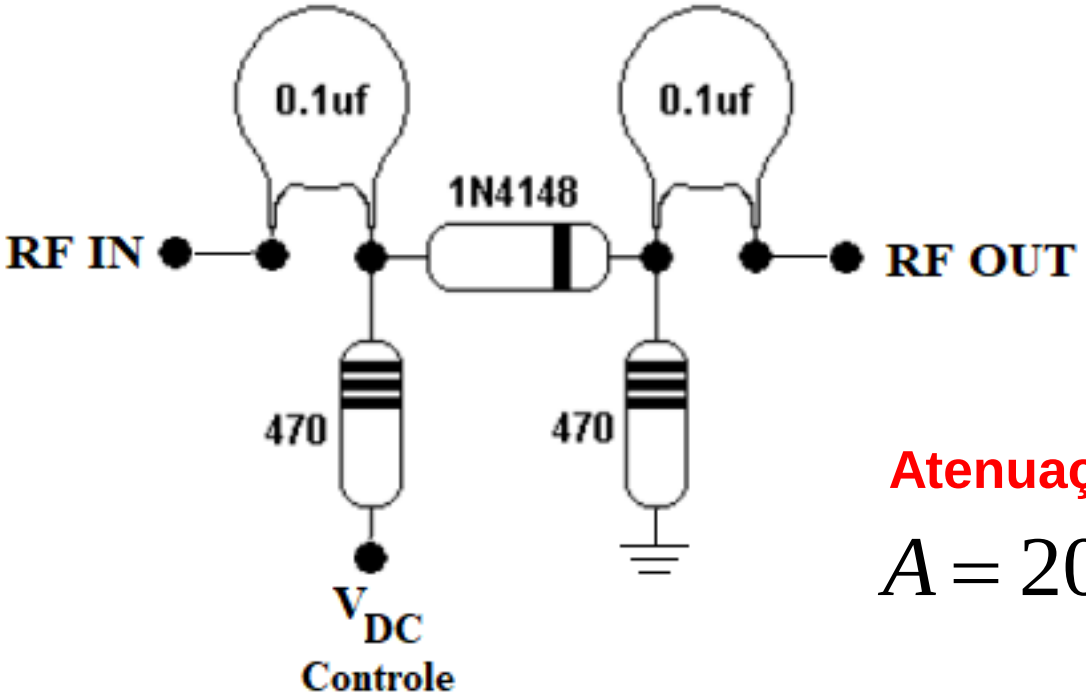


Montagem-1



Montagem-2

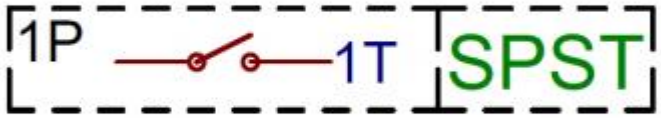
Chave de Sinais com Diodo



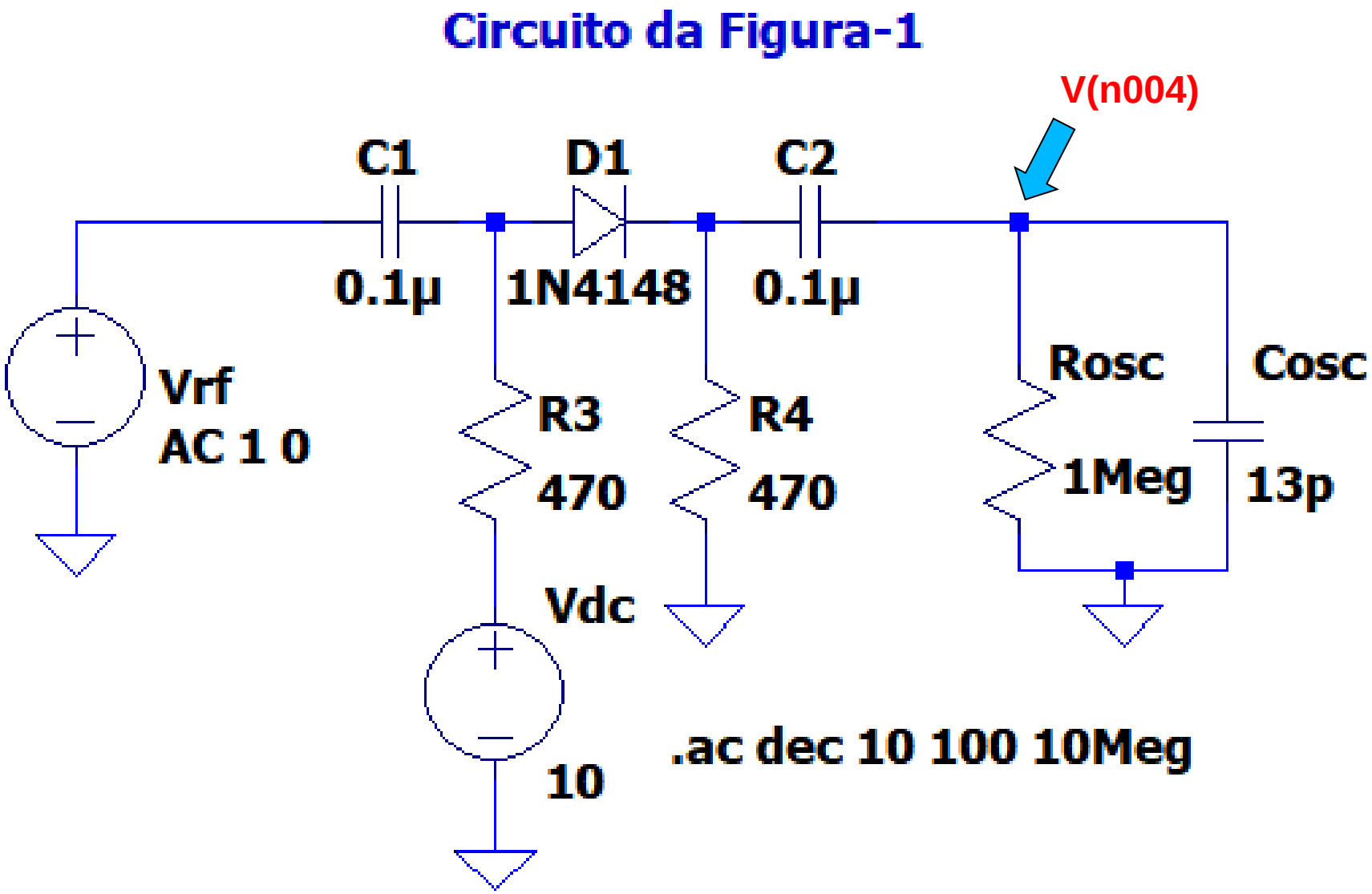
Atenuação (dB):

$$A = 20\log_{10}(V_{out-off} / V_{out-on})$$

Single Pole, Single Throw Switch

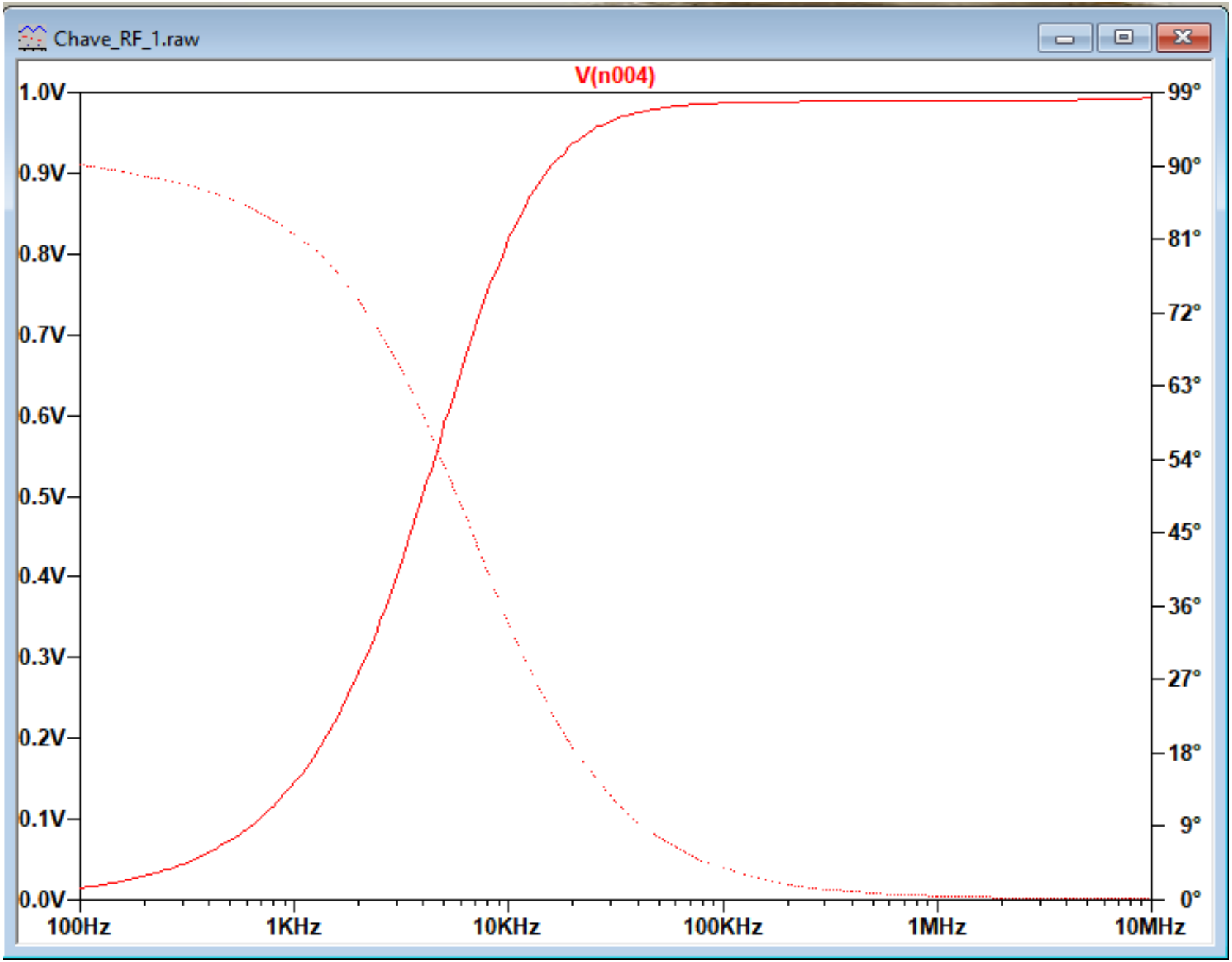


Simulação no LTspice da Chave de Sinais com Diodo



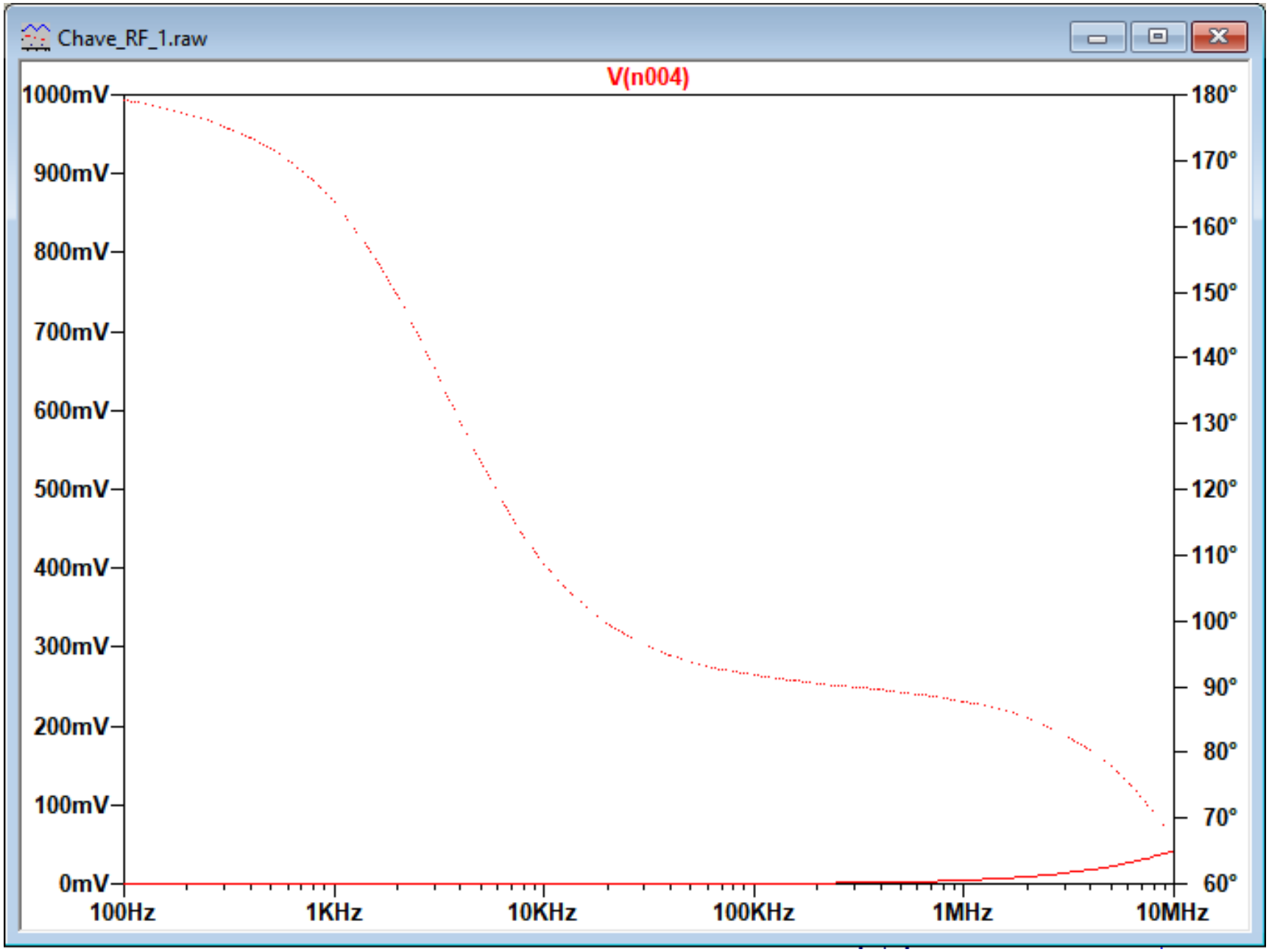
Chave_RF_1.asc

Diagrama de Bode (Log-Linear)



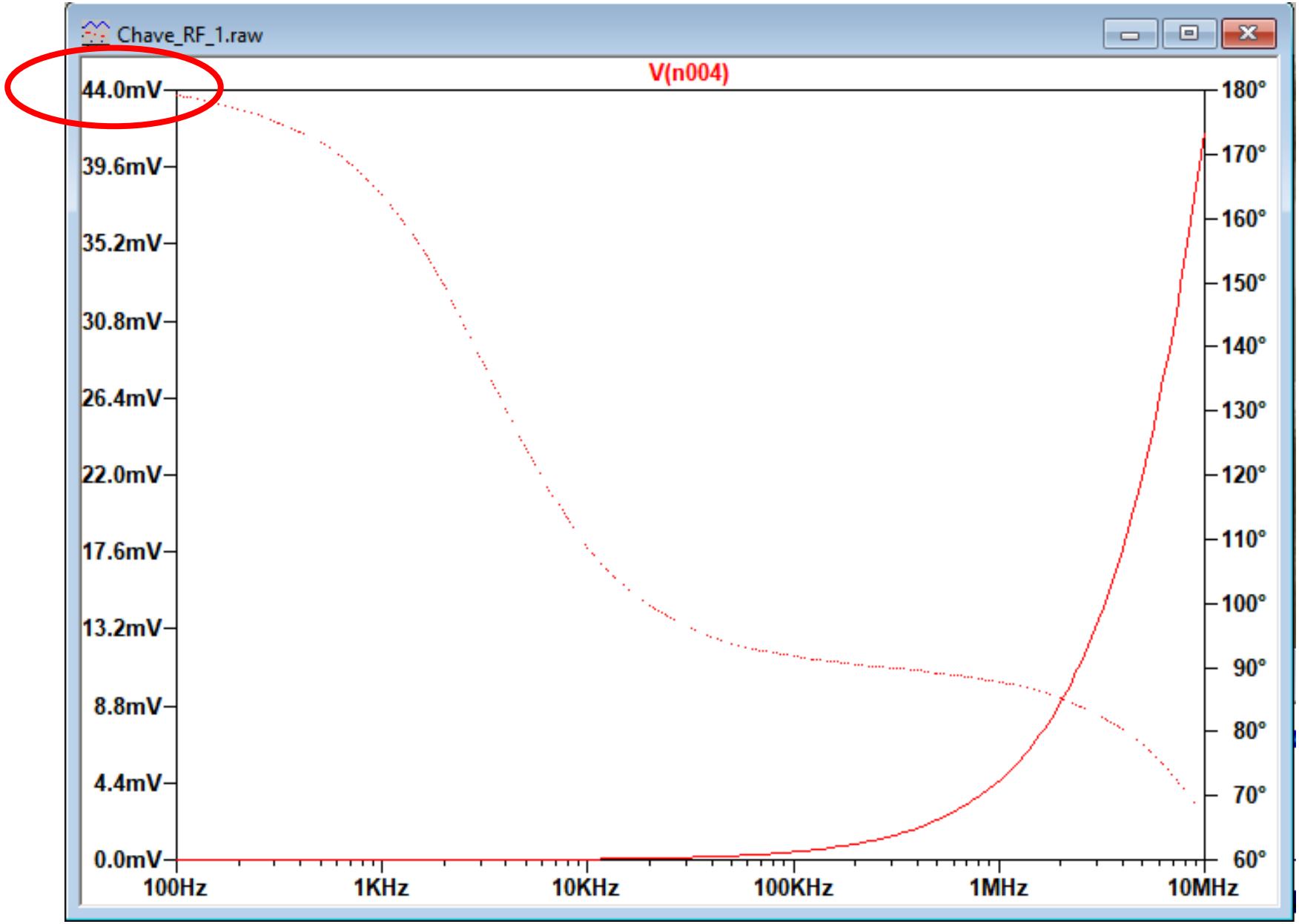
V_{DC} = +10V

Diagrama de Bode (Log-Linear)



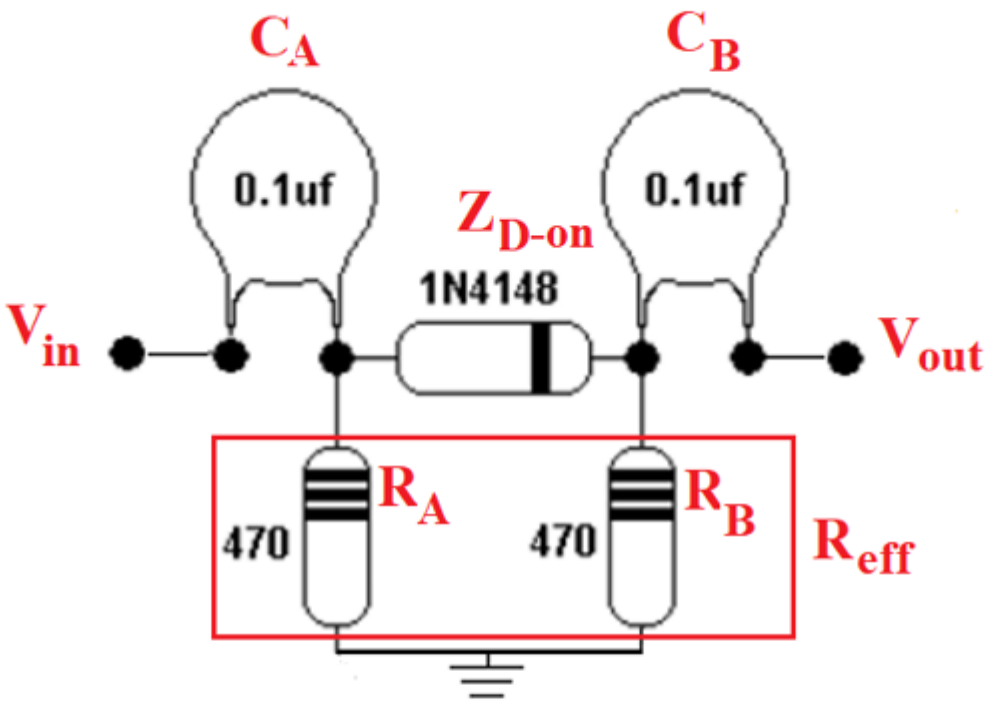
V_{DC} = -10V

Diagrama de Bode (Log-Linear)



V_{DC} = -10V

Modelo AC do Circuito da Figura-1 (V_{DC} = +10V)



$$V_A = I_A R_A \qquad V_B = I_B R_B$$

Diodo Ligado:

$$Z_{D-on} << R_A, R_B => R_{eff} \simeq R_A // R_B => R_A = R_B => R_{eff} = R_A / 2 = R_B / 2$$

$I_{out} \simeq 0$ (Alta Impedância do Osciloscópio na Saída) => $V_{out} \simeq I_{in} R_{eff}$

$$I_{in} = \frac{V_{in} - V_{out}}{Z_{C_A}} = j\omega C_A (V_{in} - V_{out})$$

$$V_{out} = j\omega R_{eff} C_A (V_{in} - V_{out})$$

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{j\omega R_{eff} C_A}{1 + j\omega R_{eff} C_A}$$

Modelo: $$H(j\omega) = \frac{V_{out}}{V_{in}} = k\frac{j\omega R_{eff} C_A}{1 + j\omega R_{eff} C_A}$$

Onde k é o fator de ganho (ou atenuação),

(Ajuste por Mínimos Quadrados, Programa: Lab_10.sce)

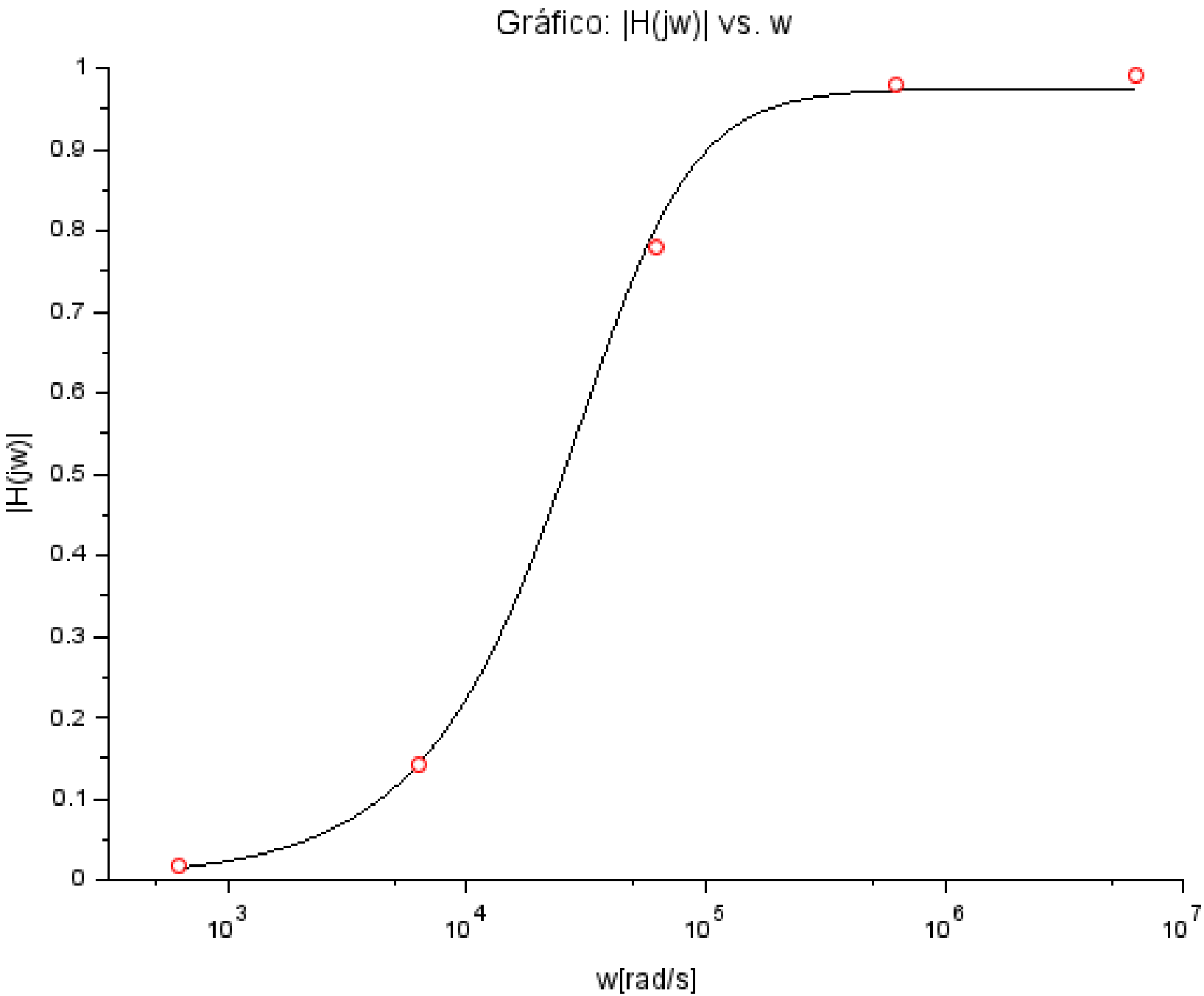
```
//Programa: Lab_10.sce
clear;
//Pontos Experimentais (Item-1, +10V)
Vin = 2; //[Vp-p]
Vout = [34.2e-3 286e-3 1.56 1.96 1.98]; //[Vp-p]
f = [100 1e3 1e4 1e5 1e6];//[Hz]
w = 2*%pi*f; //[rad/s]
Hp = Vout/Vin;

//Modelo:
C = 0.1e-6;//[F]
R = 470;//[ohms]
Reff = R/2;//[ohms]
x = abs((%i*w*C*Reff)./(1+%i*w*C*Reff)); //Função Base
y = Hp;
k = sum(y.*x)/sum(x.*x); //Ajuste da Constante do Modelo
M = 100; //Número de pontos para plotar modelo
fp = logspace(log10(100),log10(1e6),M); //[Hz]
wp = 2*%pi*fp; //[rad/s]
//Função:
Hm = k*abs((%i*wp*C*Reff)./(1+%i*wp*C*Reff));

//Gráfico: |H(jw)| Log-Linear
scf(1);
plot2d("ln",wp,Hm);
plot(w,Hp,'or');
xlabel('w[rad/s]');
ylabel('|H(jw)|');
title('Gráfico: |H(jw)| vs. w');
```

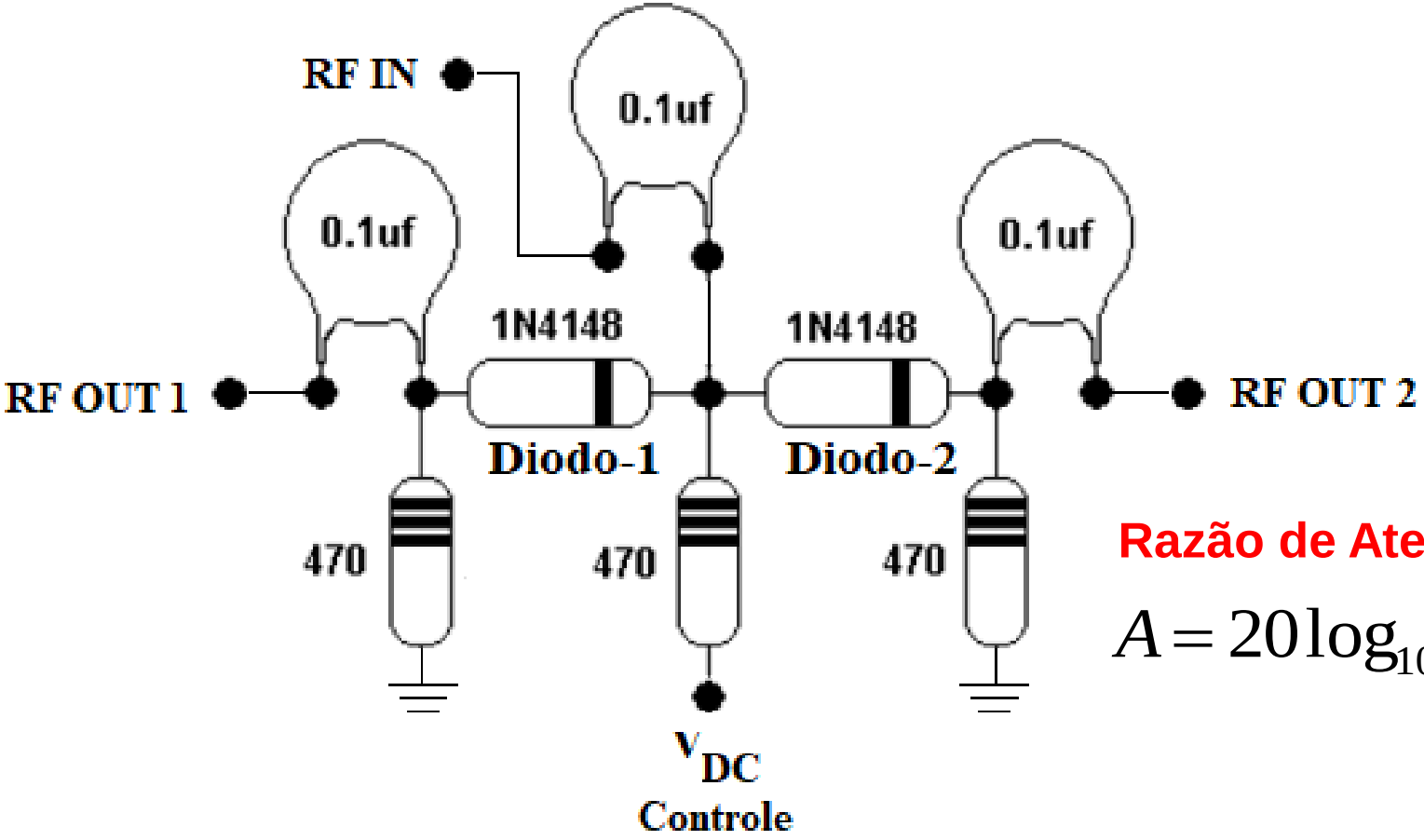
Ajuste do Modelo de H(jω)

Diagrama de Bode: Modelo Ajustado de |H(jω)|



Comutador de Sinais com Diodos

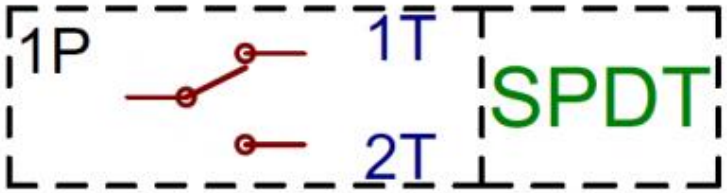
(Uma Entrada, Duas Saídas)



Razão de Atenuação (dB):

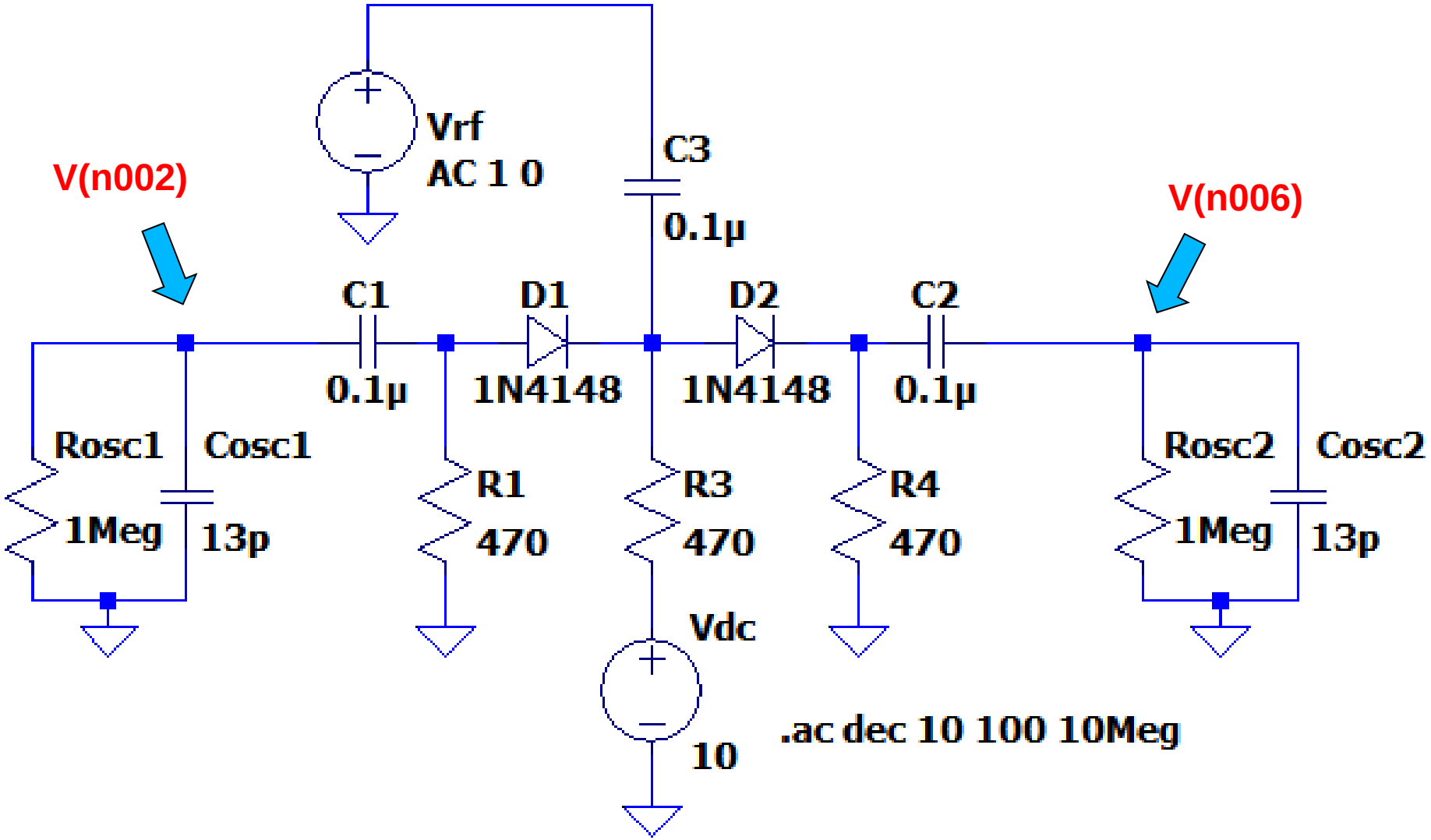
$$A = 20\log_{10}(V_{out-1}/V_{out-2})$$

Single Pole, Double Throw Switch



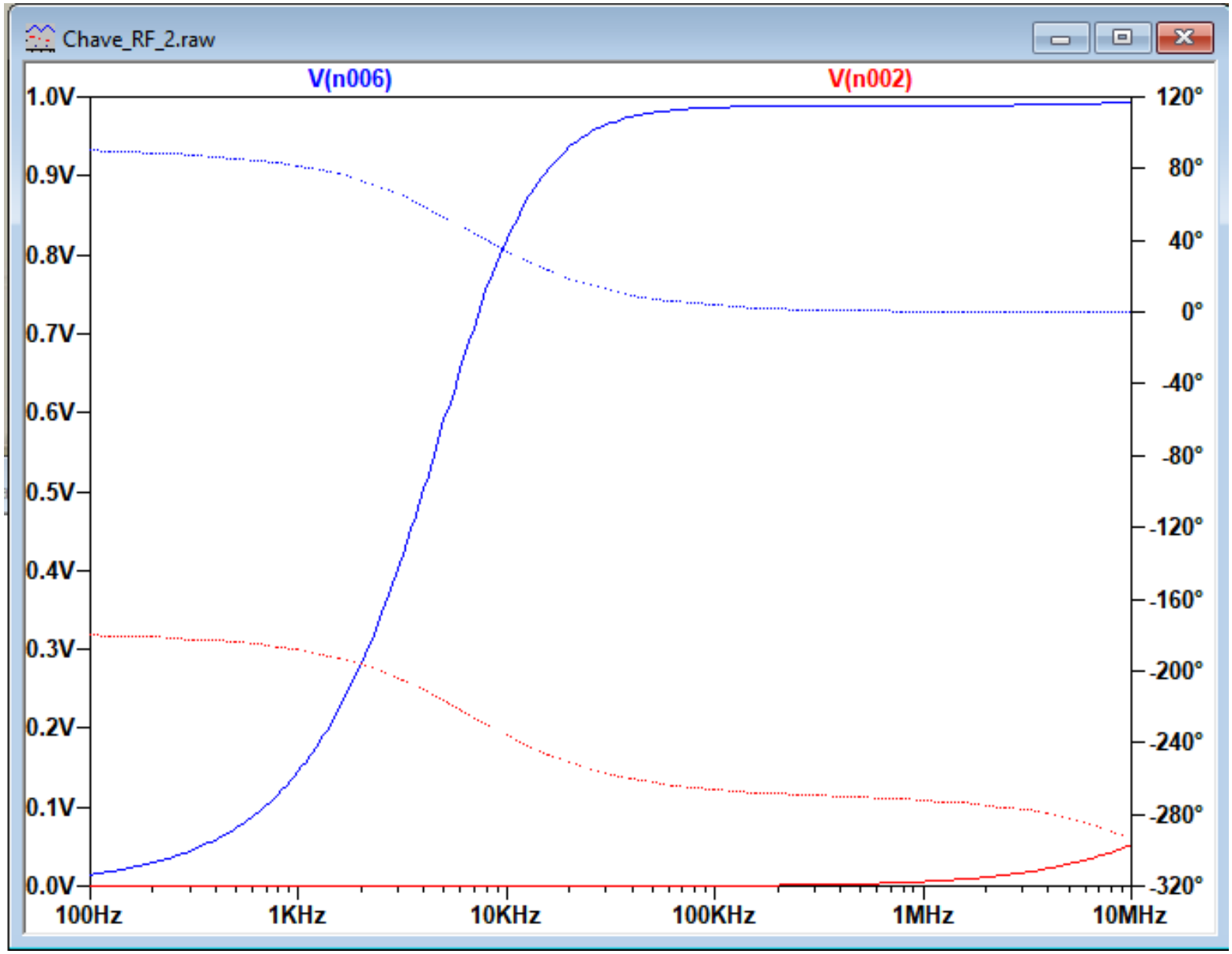
Simulação no LTspice da Comutador de Sinais com Diodos

Circuito da Figura-2



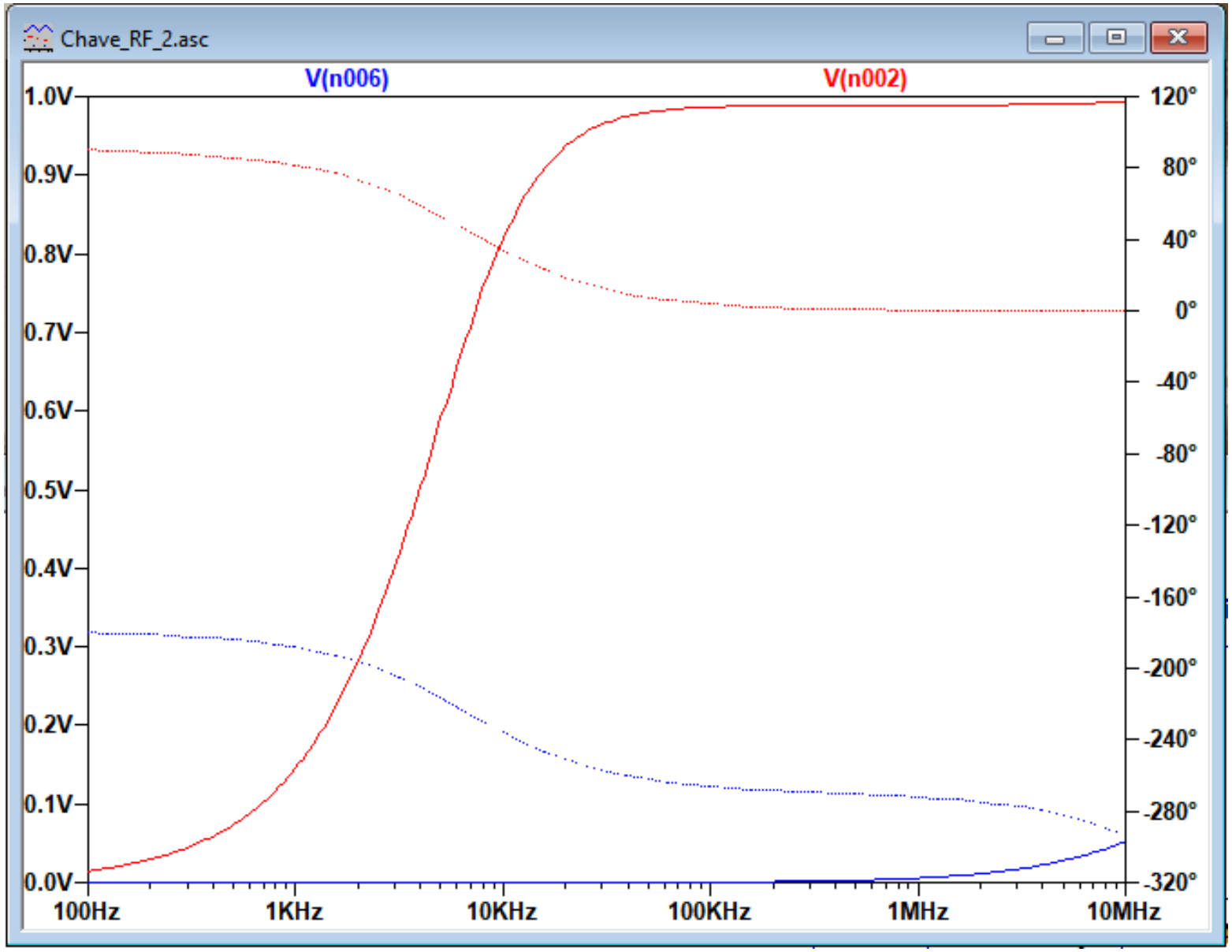
Chave_RF_2.asc

Diagrama de Bode (Log-Linear)



V_{DC} = +10V

Diagrama de Bode (Log-Linear)

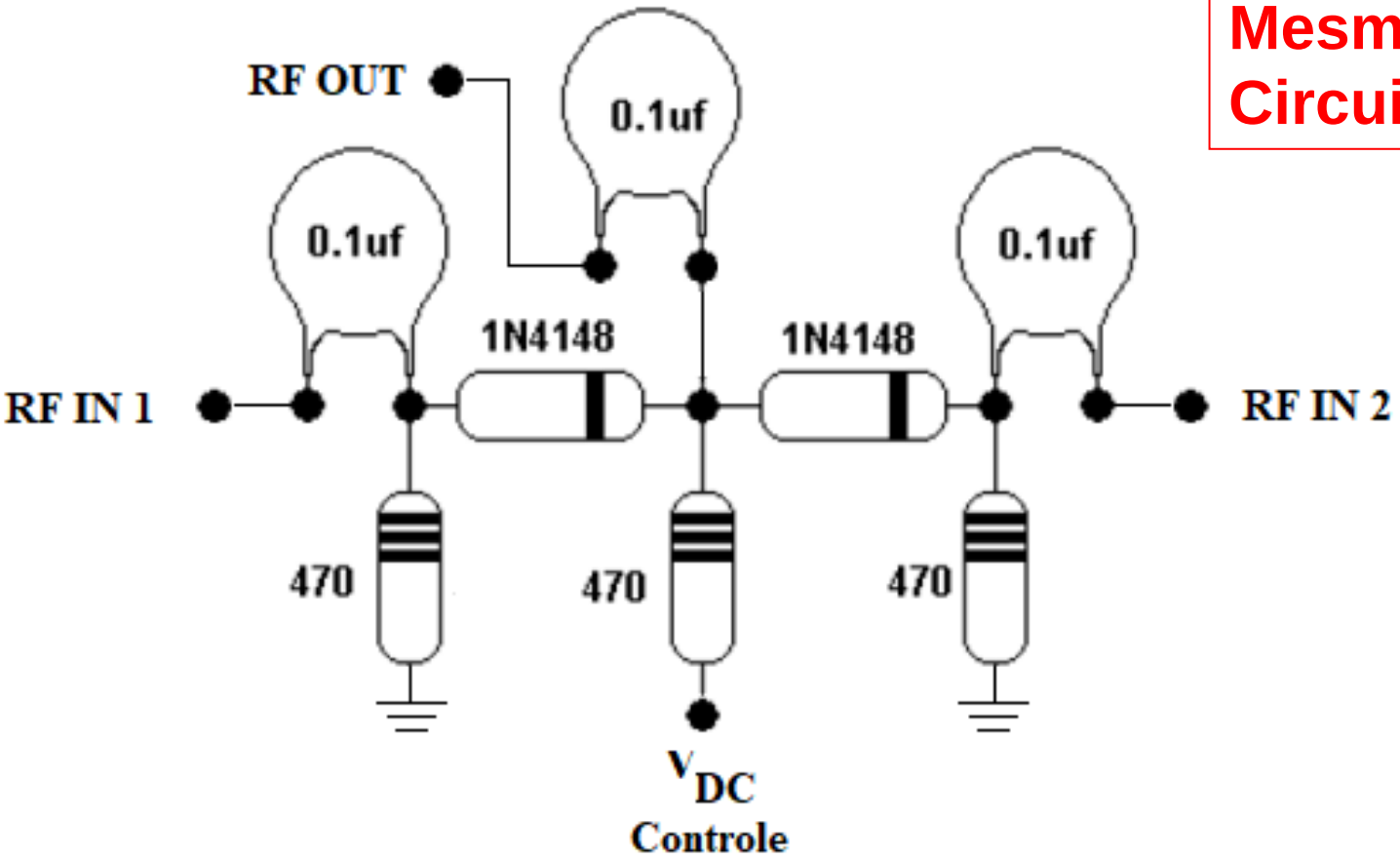


V_{DC} = -10V

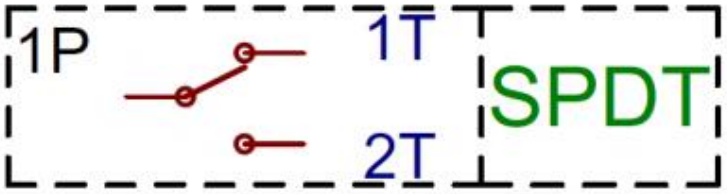
Comutador de Sinais com Diodos

(Duas Entradas, Uma Saída)

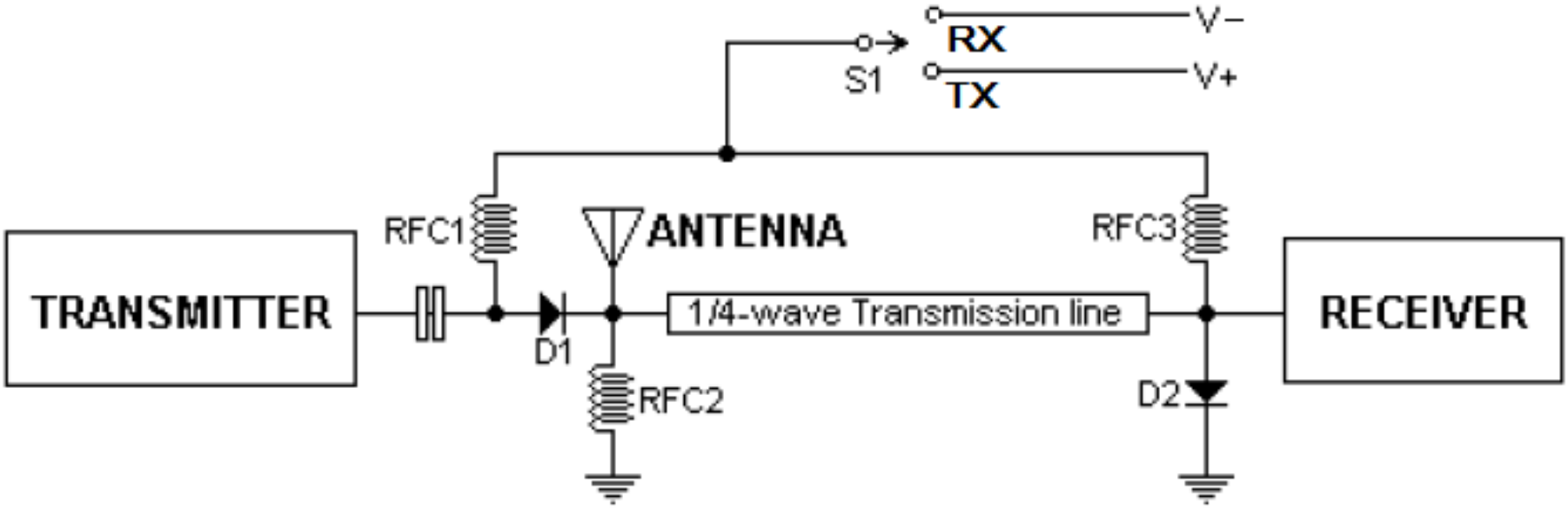
Mesmo Circuito !!!

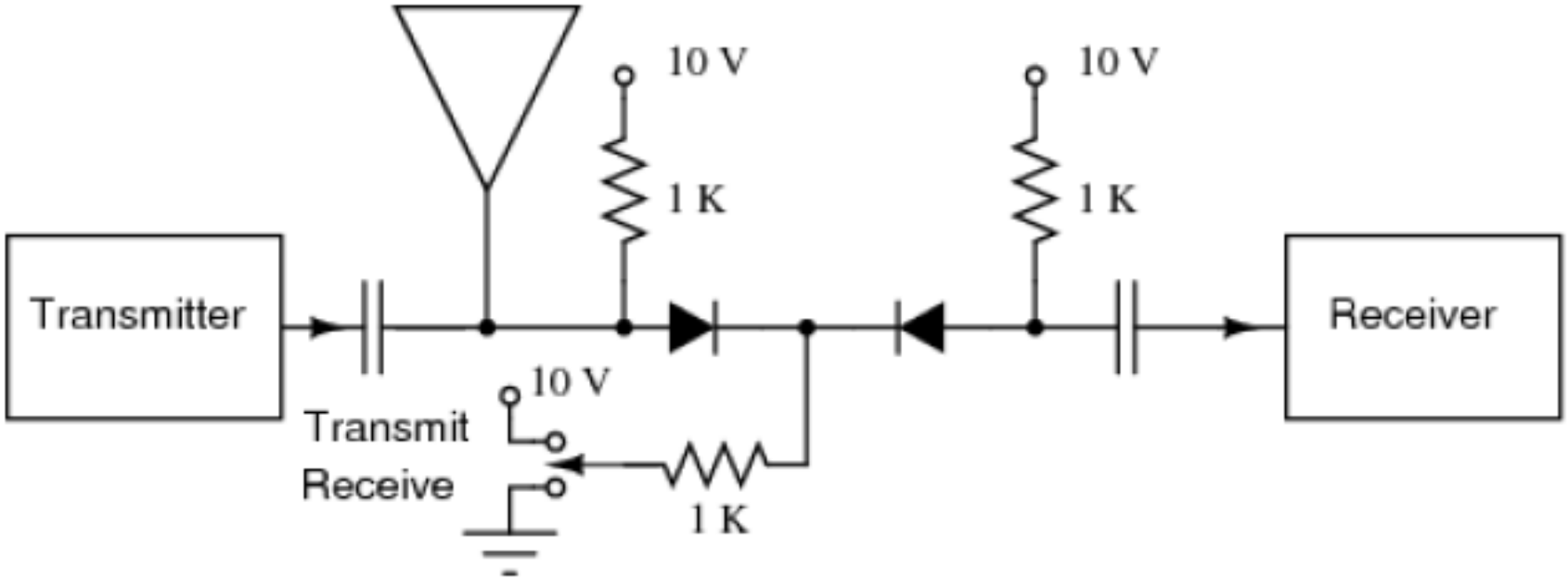


Single Pole, Double Throw Switch



Sistemas de Comunicação Half-Duplex





Comutador de Antenas

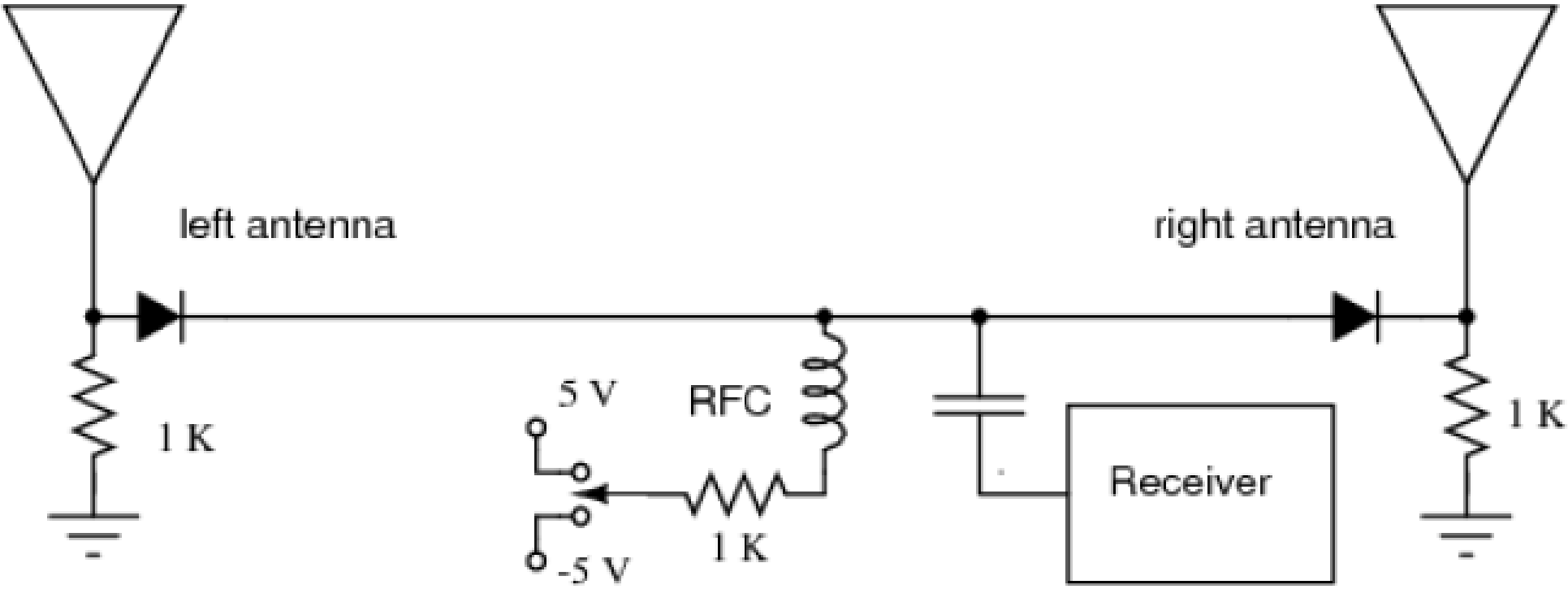
(Receptor Único)

Direcional N S

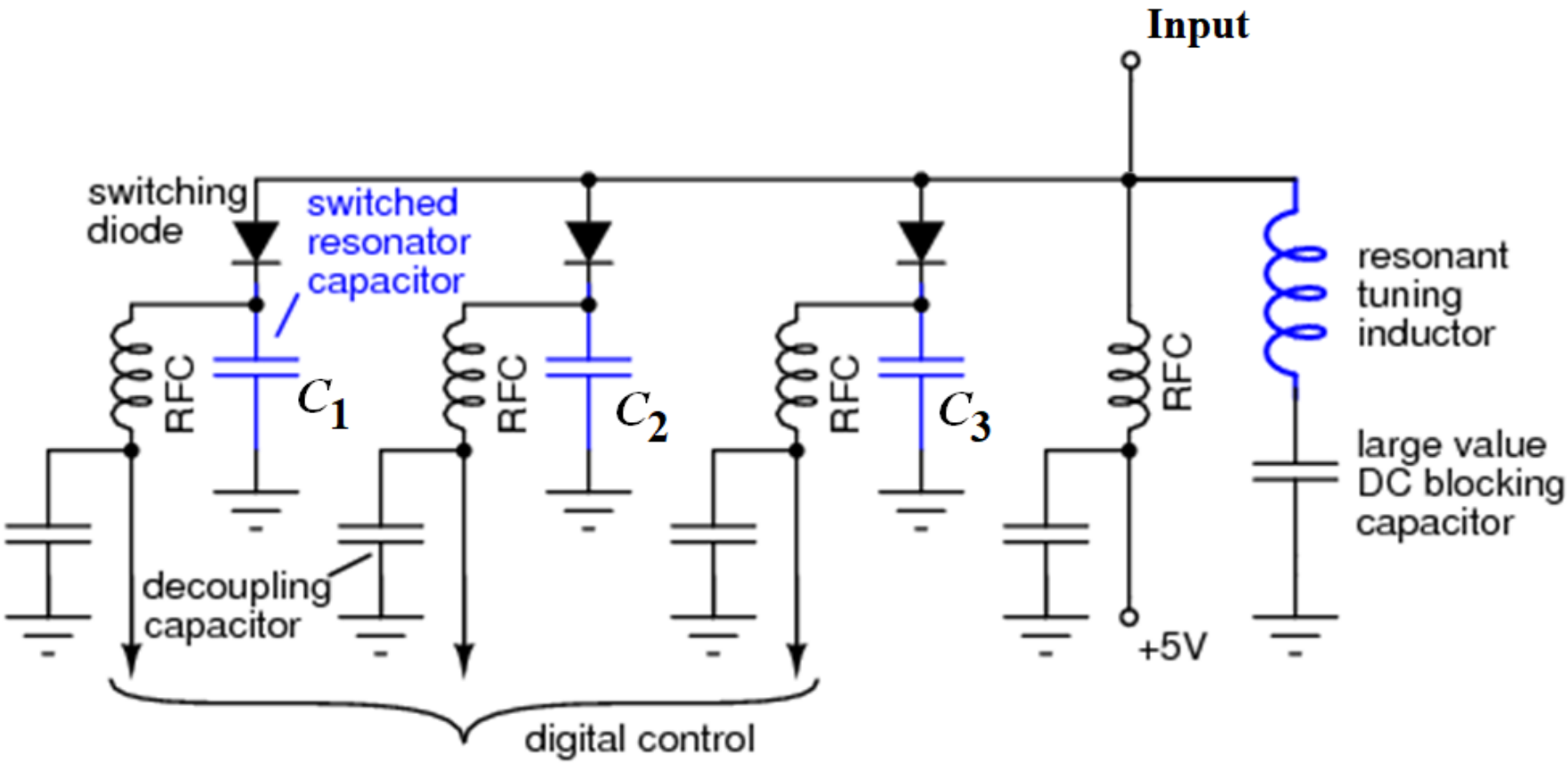
O L Direcional

Polarização Vertical

Polarização Horizontal



Filtro Ressonante Digitalmente Controlado



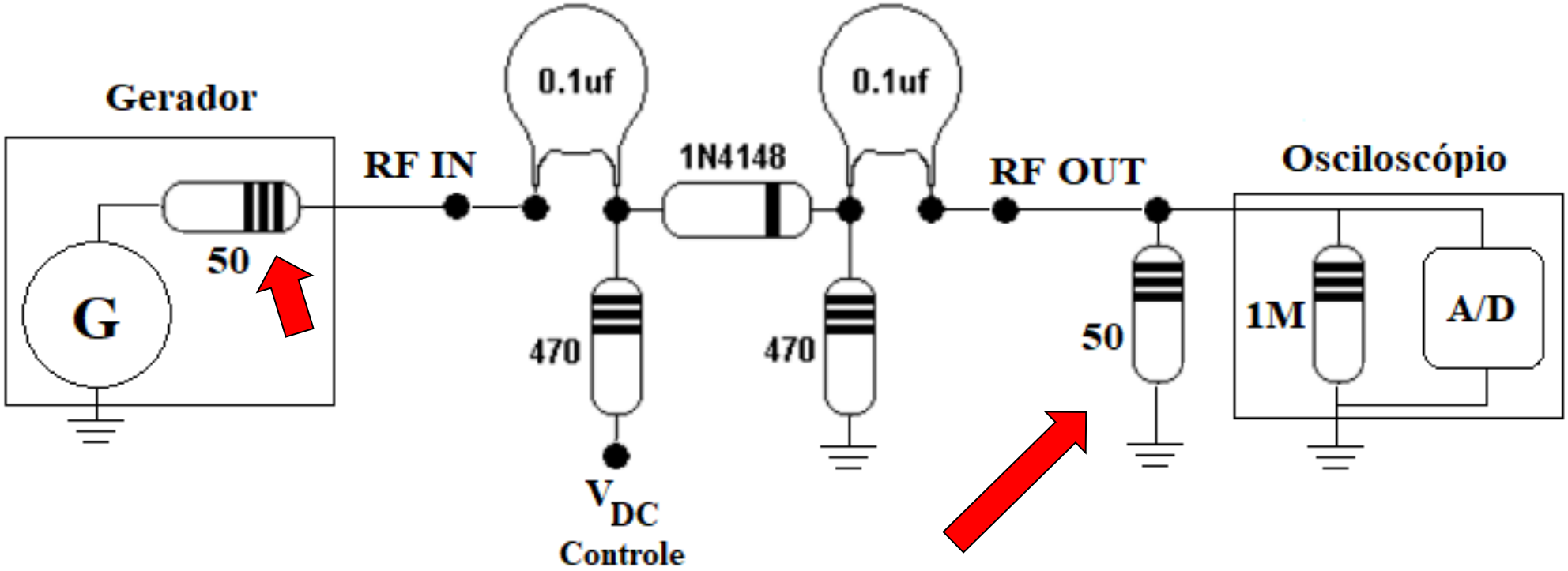
$$C_T = \sum_{i=1}^{3} s_i C_i \qquad C_i = 2^{(i-1)} C_0 \qquad f_0 = \frac{1}{2\pi\sqrt{LC_T}}$$

$$s_i = \{0,1\}$$

https://www.allaboutcircuits.com/textbook/semiconductors/chpt-3/diode-switching-circuits/

Chave de Sinais com Diodo

Acoplamento de Impedâncias para alta-frequência.

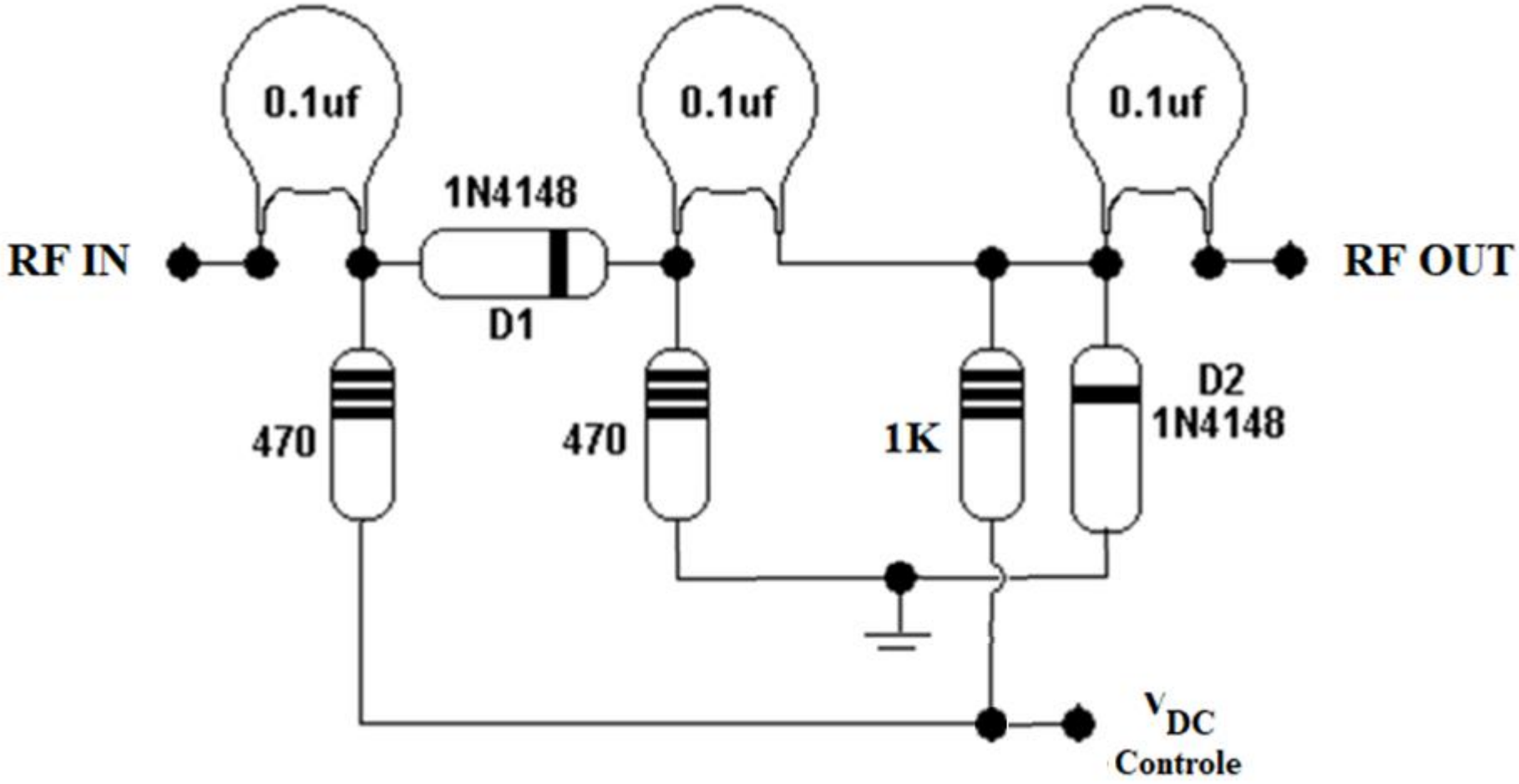


Incluir Resistor de 50Ω

Condição de Máxima Transferência de Potência do Sinal

Aumento da Razão Sinal / Ruído (*Signal to Noise Ratio* – SNR)

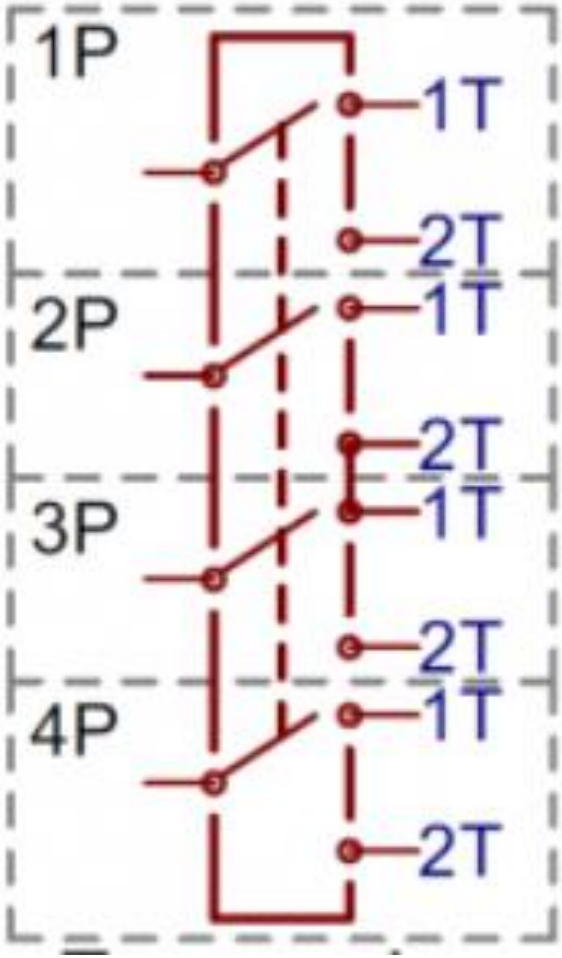
Chave de Sinais com dois Diodos
(Configuração Série-Paralelo)



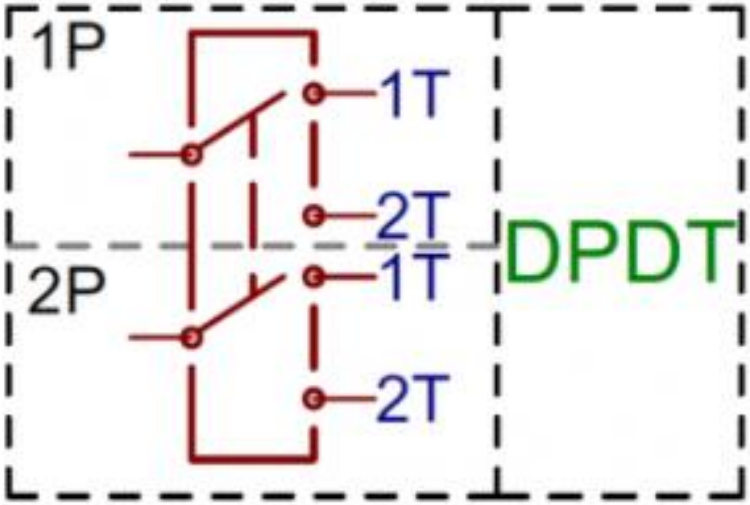
D1 – Série
D2 – Paralelo

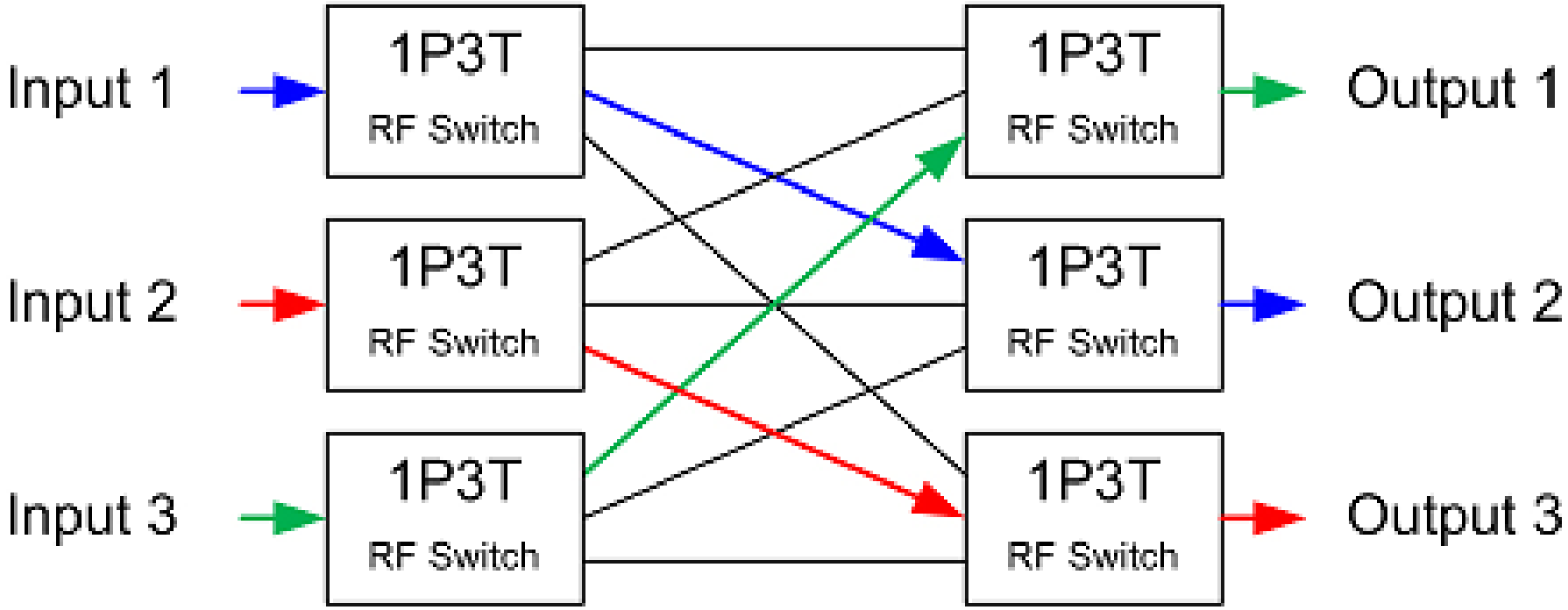
Outros Comutadores de Sinais com Diodos

Four Pole, Double Throw Switch (4PDT)



Double Pole, Double Throw Switch (DPDT)



Matriz Comutadora de Sinais com Diodos

N = 3 Entradas, M = 3 Saídas